Codefeast - 6

1. Ideal Read.

For Ideal Case, Row 1 & Col 1 are grounded.

$$I_{col0} = (V\_row0 - V\_col0) / R[0][0]$$

$$= (1V - 0V) / 1k\Omega = 1mA$$

2. Sneak Path Read

Row 1 & Col 1 are floating.

$V\_row1 = x$ $\quad$ $V\_row0 = 1V$

$V\_col1 = y.$ $\quad$ $V\_col0 = 0V$

KCL @ row1

$$(x-0)/2k\Omega + (x-y)/1k\Omega = 0$$

x'ly by $2k\Omega$

$$x + 2x - 2y = 0$$
$$3x - 2y = 0$$
$$y = 1.5x$$

KCL @ col1

$$(y-1)/2k\Omega + (y-x)/1k\Omega = 0$$

x'ly by $2k\Omega$

$$y - 1 + 2y - 2x = 0$$
$$3y - 2x = 1$$

Sub $y = 1.5x$

$$\Rightarrow 3(1.5)x - 2x = 1$$
$$2.5x = 1$$
$$x = 0.4V$$

$V\_row1 = 0.4V$

$V\_col1 = 0.6V$

3. Actual I_col0 including Sneak path

Current through Intended path R[0][0]

$$I_{intended} = (1V - 0V)/1k\Omega = 1mA$$

Sneak path current entering col0 through R[1][0]:

$$I_{sneak} = (V\_row1 - V_{col0}) / R[1][0]$$

$$= (0.4V - 0V)/2k\Omega = 0.2mA.$$

Total Sensed Coloumn Current.

$$I\_col0\_actual = I_{intended} + I_{sneak}.$$
$$= 1mA + 0.2mA.$$
$$= 1.2mA.$$

Actual $I\_col0 = 1.2\,mA$.

4 Explanation.

The ideal MVM result expects only the current from the selected cell R[0][0], which is 1mA. However, because row1 & col1 are floating, an additional sneak path current of 0.2 mA flows into col0, increasing the sensed current to 1.2mA.

This corrupts the MVM result because the coloumn current no longer represents only the intended dot product contributio

In large cross bar array's, many such sneak paths can exists, causing read errors & making accurate sensing more difficult.